

FORMULA DOCUMENTATION

1. Empirical Probability Formula

Empirical probability is used to calculate probability using actual observed data from the dataset.

Formula:

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

Where:

- Favorable Outcomes = number of desired outcomes
- Total Outcomes = total observations in dataset

Purpose:

This formula is used to calculate probabilities such as:

- probability of passing,
- probability of high attendance,
- probability of participation in group discussion.

2. Conditional Probability Formula

Conditional probability is used to calculate the probability of one event occurring when another event has already occurred.

Formula:



Where:

- A = first event
- B = second event

- $A \cap B$ = common occurrence of both events

Purpose:

Used to determine:

- probability of passing given participation in group discussion.
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3. Mean Formula of Binomial Distribution

Mean represents the average expected value.

Formula:

$$\mu = np$$

Where:

- n = number of trials
- p = probability of success

Purpose:

Used to calculate the average expected number of students passing among selected students.

4. Variance Formula of Binomial Distribution

Variance measures spread or variability in data.

Formula:

$$\sigma^2 = np(1 - p)$$

Where:

- n = number of trials
- p = probability of success

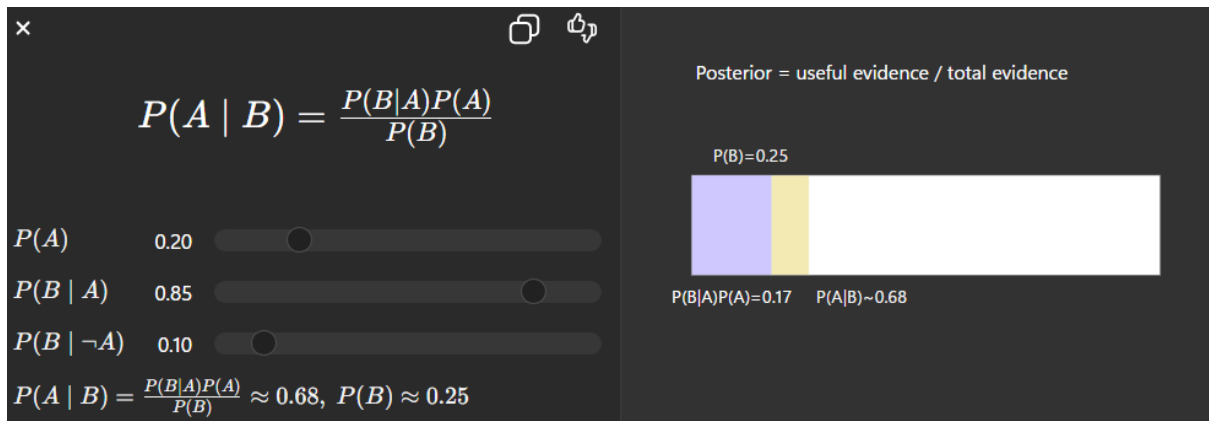
Purpose:

Used to determine variation in passing probabilities.

5. Bayes Theorem Formula

Bayes theorem is used to calculate conditional probability using prior probabilities.

Formula:



Where:

- A = event of passing exam
- B = event of high attendance

Purpose:

Used to predict passing probability based on attendance.

6. Intersection Formula

Intersection represents common elements between sets.

Formula:

$$A \cap B$$

Where:

- A = students with study hours > 10
- B = students with attendance > 80

Purpose:

Used in Venn Diagram to show students satisfying both conditions.

GRAPH DOCUMENTATION

1. Countplot of Final Exam Pass

Description

A countplot is used to display the frequency of categorical data.

In this project, the countplot shows:

- number of students who passed,

- number of students who failed.

Purpose

This graph helps compare pass and fail counts visually.

Observation

The graph shows whether passing students are greater or fewer than failing students.

2. Histogram of Attendance

Description

A histogram is used to display the distribution of numerical data.

This histogram represents attendance percentages of students.

Purpose

It helps understand:

- attendance spread,
- frequency distribution,
- common attendance ranges.

Observation

Most students are concentrated around certain attendance values.

3. Histogram of Study Hours

Description

This histogram represents distribution of student study hours.

Purpose

Used to analyze:

- study habits,
- frequency of study hours,
- concentration of students in different ranges.

Observation

The graph shows how many students study for specific hour ranges.

4. Scatter Plot of Study Hours vs Previous Test Score

Description

A scatter plot is used to show relationship between two numerical variables.

This graph compares:

- study hours
- previous test scores

Purpose

Used to determine whether study hours affect test scores.

Observation

Students with higher study hours generally have better test scores.

5. Pie Chart of Final Exam Pass

Description

A pie chart represents proportions of categories as slices of a circle.

This pie chart shows:

- percentage of passed students,
- percentage of failed students.

Purpose

Used for easy visual comparison of pass and fail percentages.

Observation

The larger slice represents the category with higher percentage.

6. Bar Graph of Group Discussion

Description

A bar graph displays categorical data using rectangular bars.

This graph shows:

- students participating in group discussion,
- students not participating.

Purpose

Used to compare participation levels.

Observation

The graph clearly shows which category has more students.

7. Box Plot of Attendance

Description

A box plot represents spread and distribution of data.

It shows:

- minimum value,
- maximum value,
- median,
- quartiles,
- outliers.

Purpose

Used to analyze variation in attendance data.

Observation

The box plot helps identify spread and unusual attendance values.

8. Heatmap

Description

A heatmap visually represents correlation between numerical variables using colors.

Purpose

Used to analyze relationships among:

- study hours,
- attendance,
- previous test scores.

Observation

Darker colors indicate stronger relationships between variables.